Colt Goodrum, Julian Bernabe-Hernandez, Gui Panzi, Tiago Field, Addison Richardson

Schuckers/Ageenko

DTSC 2302

28 April 2026

Violent and Disruptive Incidents in NYC Report

Introduction

School safety is an ongoing concern that applies not only to educators and other education workers but also to decision-makers concerned with school safety issues. With the increasing diversity of student populations in schools, identifying the variables that contribute to the probability of a discipline problem being brought to the attention of officials becomes more and more important. The purpose of this study is to identify if school-level demographic variables like total enrollment, race, special education enrollment, and male enrollment can be used to predict safety cases.

This is definitely a very critical topic to pursue, considering the impact of such an association on the development of educational policy. The possibility to identify what type of schools would more likely experience such incidents will make it possible to prepare for these events beforehand. One thing to remember is that even though there are correlations between the variables, this investigation has to be conducted carefully because these correlations do not imply any innate traits of students.

Two different types of datasets at the level of schools were used to figure this out. The first one is the longitudinal demographics dataset, and the second one is the annual incidents in dozens of categories, from simple conflicts to severe violations of rules, like carrying weapons and assault, are measured by a single variable named Total Incident Count. Two models were developed with one being a regression model for estimating the total number of incidents and a classification model.

Based on the results, the conclusion can be made that the main feature is the number of students in the school, followed by demographic features such as the percent of special needs students and the race category. The Random Forest technique outperformed other methods in terms of regression and classification, including Linear Regression and k-Nearest Neighbors. For regression, Random Forest achieved an R^2 of 0.52, while for classification, the accuracy was 79.1%. Therefore, the results suggest that there is some signal in demographic features, although as there is still some variance unexplained, others must be taken into account.

The remaining sections of the paper are organized as follows. First, the Regression Model section describes the three types of regression models and analyzes how effective these regression models were at predicting total incidences. The Classification Model section discusses how the schools were categorized and evaluates the three types of classification models used. And the Conclusion section summarizes the conclusions that were made.

Regression Model

To predict school level total incident counts, multiple regression models were made using demographic variables as predictors, including total enrollment, special ED percentage, english language learner, racial composition percentages, and male percentages. Three different regression models were tested: Linear Regression, Random Forest Regression, and k-Nearest Neighbors Regression. The data was cleaned by converting feature columns to numeric values, and removing rows and columns with missing values. The dataset was then split into training and testing sets to evaluate generalization performance.

The Linear Regression model serves as a baseline since it's the most simple and interpretable. However, it assumes a linear relationship between demographic features and incident counts. To improve performance, a Random Forest Regressor was made. This model builds multiple decision trees and averages their outputs, allowing it to capture nonlinear relationships and interactions between variables. A k-Nearest Neighbors Regressor was also tested as a distance based, and non parametric comparison model.

Model performance was assessed using the R^2 value. Among the three models, the Random Forest Regressor had the highest R^2 value, showing us that it’s the best fit for the data. Linear Regression performed moderately well, but was limited by the assumption of linearity, while kNN performed the weakest, likely due to its sensitivity to feature scaling and high dimensional data. Overall, the Random Forest model was selected as the final regression model due to its better accuracy and ability to model complex, nonlinear relationships between demographic factors and school incident rates. However, even though it has better performance, the model still explains only a part of the variance, suggesting that additional factors beyond demographics influence incident frequency.

Classification Model

The goal of the classification model was to predict whether a school-year record would have high incident levels or low incident levels based on demographic and enrollment characteristics. To build the dataset, school demographic data was merged with incident report data after cleaning and standardizing school names and school year formats. Incident counts from multiple categories were converted to numeric form and combined into a single variable called “Total_Incidents”. A binary target variable was then created by labeling schools as “high incidents” if their total incidents were above the median and “low incidents” if the total was below or at the median.

Several demographic predictors were used, including total enrollment and percentages of different student groups (ELL, SPED, racial groups, and male enrollment). The data was split into training and testing sets, the 3 models were trained and compared: Logistic Regression, Random Forest and k-Nearest Neighbors (kNN). Each model was evaluated using accuracy on the test set, and Random Forest was treated as the strongest model overall due to its ability to capture nonlinear patterns.

Model performance was further assessed using a confusion matrix, which showed how well the model correctly identified high-incident and low-incident schools. Overall, the Random Forest classifier performed the best and demonstrated that demographic and enrollment factors can provide useful information for identifying high risk & low risk schools. However, the model is limited by the use of a median cutoff and lack of more predictors such as location, funding and staffing, which could improve accuracy.

Conclusion

This project demonstrated that demographic and enrollment variables carry real predictive signals for school safety outcomes in NYC public schools, but that the relationship is more nuanced than any demographic model alone can fully explain. Across both the regression and classification tasks, the Random Forest method consistently outperformed Linear Regression and kNN, achieving an R^2 of 0.52 and a classification accuracy of 79.1% respectively. These results confirm that school size and demographic composition meaningfully predict incident volume and risk level, but they also leave substantial variance unexplained, which is itself an important finding.

Total enrollment being the strongest predictor in both models can be attributed to the fact that larger schools have more students, more daily interactions, and more opportunities for incidents to occur and be reported. The significance of special education percentage likely reflects the additional documentation and behavioral reporting requirements associated with SPED populations rather than any inherent characteristic of those students, which is a distinction that matters greatly when interpreting these results. The role of racial composition percentages as predictors is harder to interpret cleanly and deserves careful consideration as these correlations almost certainly reflect the downstream effects of systemic inequities in school funding, neighborhood resources, and historical policy rather than anything intrinsic to the student populations themselves.

This brings up the most significant limitation of the model and its most important ethical implication. A model trained on demographic data to flag "high-risk" schools risks encoding and reinforcing existing inequities rather than explaining them. If such a model were applied in practice to allocate resources or interventions, it could effectively penalize schools serving already disadvantaged populations without addressing the underlying structural conditions driving those outcomes. The features missing from this analysis could be but are not limited to school funding, teacher-to-student ratios, neighborhood crime rates, access to mental health services. These are the variables that would help distinguish between a school that is high-risk because of its demographics and one that is high-risk because it is under-resourced. Without them, the model describes a pattern but cannot responsibly prescribe a response, especially given the sensitive nature of the issue as well as the current standing of the education system.

Despite these limitations, the models perform well as an exploratory tool. The consistency of feature importance rankings across both the regression and classification models strengthens confidence that enrollment and demographic composition are genuine signals rather than noise. Future work that incorporates resource and geographic variables, treats incident categories separately rather than as a single total, and uses more longitudinal modeling to track individual schools over time would substantially improve both predictive power and interpretive value.

Appendix 1

I acknowledge the use of Copilot in the completion of this assignment. The Copilot was used in the following ways in this assignment: merging the datasets.

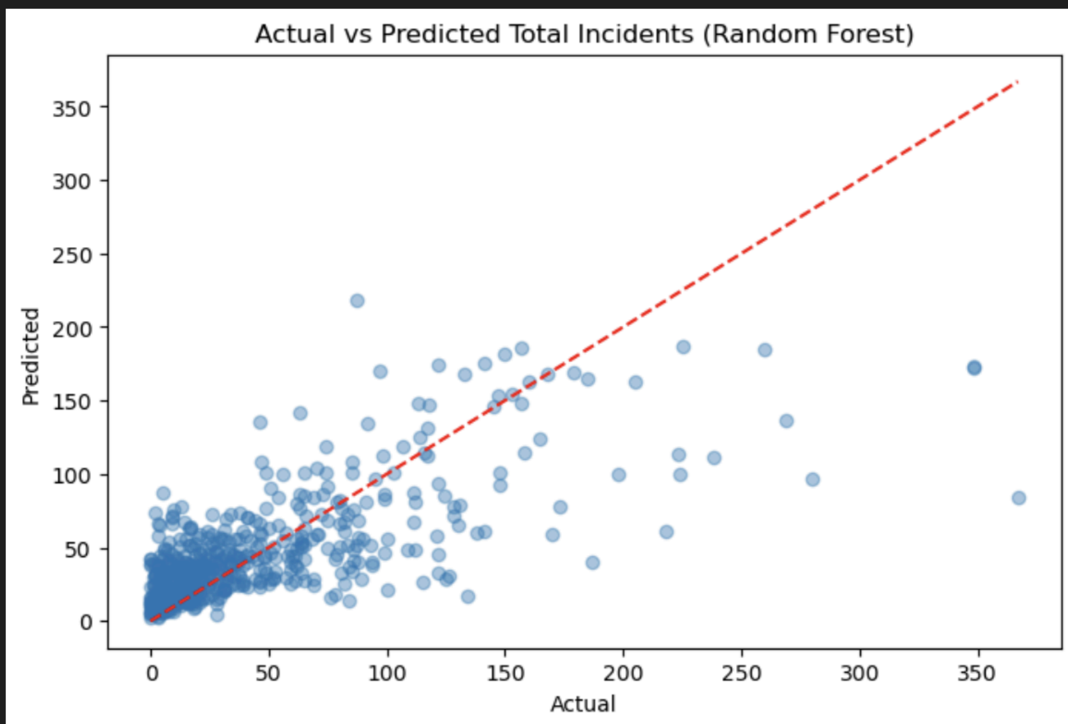
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.neighbors import KNeighborsRegressor, KNeighborsClassifier
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, confusion_matrix

demographics_df = pd.read_csv('https://docs.google.com/spreadsheets/d/e/2PACX-1vRSnla2AtvwQL1VIdBMtScWgb7HOfT3PfNTSKEbyYbweLV6oI9tSlLgMlWw37x31WBxGQePIiF_d5yi/pub?output=csv')
incidents_df = pd.read_csv('https://docs.google.com/spreadsheets/d/e/2PACX-1vSPeJtPExJ-z5UdNprXAJrFG9PIRkltJGDzuDC2cQ9nOIUD73pp5MZjgWDvT5MPq8RcZMTTrRtLeFin/pub?output=csv')

demographics_df['Name_clean'] = demographics_df['Name'].str.upper().str.strip()
incidents_df['School Name_clean'] = incidents_df['School Name'].str.upper().str.strip()

def format_year(y):
    y_str = str(y)
    if len(y_str) == 8:
        return f"{y_str[:4]}-{y_str[4:]}"
    return y_str

demographics_df['School Year_clean'] = demographics_df['schoolyear'].apply(format_year)
incidents_df['School Year_clean'] = incidents_df['School Year'].str.strip()

incident_cols = [
    'Homicide With Weapon(s)', 'Homicide Without Weapon(s)',
    'Forcible Sex Offenses With Weapon(s)', 'Forcible Sex Offenses Without Weapon(s)',
    'Other Sex Offenses With Weapon(s)', 'Other Sex Offenses Without Weapon(s)',
    'Robbery With Weapon(s)', 'Robbery Without Weapon(s)',
    'Assault With Serious Physical Injury With Weapon(s)',
    'Assault With Serious Physical Injury Without Weapon(s)',
    'Arson With Weapon(s)', 'Kidnapping With Weapon(s)',
    'Kidnapping Without Weapon(s)', 'Assault with Physical Injury With Weapon(s)',
    'Assault with Physical Injury Without Weapon(s)',
    'Reckless Endangerment With Weapon(s)', 'Reckless Endangerment Without Weapon(s)',
    'Minor Altercations With Weapon(s)', 'Minor Altercations Without Weapon(s)',
    'Intimidation, Harassment, Menacing, or Bullying With Weapon(s)',
    'Intimidation, Harassment, Menacing, or Bullying Without Weapon(s)',
    'Burglary With Weapon(s)', 'Burglary Without Weapon(s)',
    'Criminal Mischief With Weapon(s)', 'Criminal Mischief Without Weapon(s)',
    'Larceny, or Other Theft With Weapon(s)', 'Larceny, or Other Theft Without Weapon(s)',
    'Bomb Threat', 'False Alarm', 'Riot With Weapon(s)', 'Riot Without Weapon(s)',
    'Weapon PossessionThrough Screening', 'Weapon Possession Under Other Circumstances',
    'Drug Possession', 'Alcohol Possession', 'Other Disruptive'
]
```

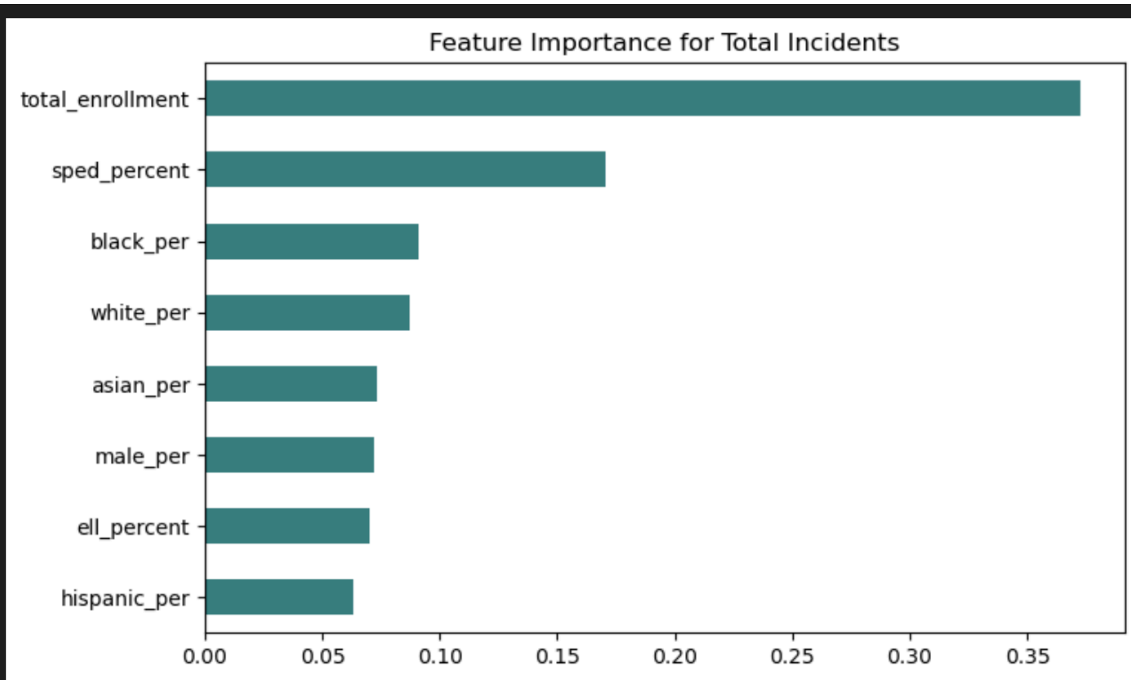
```python
for col in incident_cols:
    incidents_df[col] = pd.to_numeric(incidents_df[col], errors='coerce').fillna(0)

incidents_df['Total_Incidents'] = incidents_df[incident_cols].sum(axis=1)

merged_df = pd.merge(
    demographics_df,
    incidents_df,
    left_on=['Name_clean', 'School Year_clean'],
    right_on=['School Name_clean', 'School Year_clean'],
    how='inner'
)

features = [
    'total_enrollment', 'ell_percent', 'sped_percent',
    'asian_per', 'black_per', 'hispanic_per', 'white_per', 'male_per'
]

for feat in features:
    merged_df[feat] = pd.to_numeric(merged_df[feat], errors='coerce')

model_data = merged_df[features + ['Total_Incidents']].dropna()

X = model_data[features]
y_reg = model_data['Total_Incidents']
X_train, X_test, y_train_reg, y_test_reg = train_test_split(X, y_reg, test_size=0.2, random_state=42)

# Method 1: Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train_reg)
y_pred_lr = lr.predict(X_test)

# Method 2: Random Forest Regressor
rf_reg = RandomForestRegressor(n_estimators=100, random_state=42)
rf_reg.fit(X_train, y_train_reg)
y_pred_rf_reg = rf_reg.predict(X_test)

# Method 3: k-Nearest Neighbors Regressor
knn_reg = KNeighborsRegressor(n_neighbors=5)
knn_reg.fit(X_train, y_train_reg)
y_pred_knn_reg = knn_reg.predict(X_test)

# Evaluate Regression
print("Regression Results (R2 Score):")
print(f"Linear Regression: {r2_score(y_test_reg, y_pred_lr):.4f}")
print(f"Random Forest:     {r2_score(y_test_reg, y_pred_rf_reg):.4f}")
print(f"kNN Regressor:     {r2_score(y_test_reg, y_pred_knn_reg):.4f}")

median_inc = model_data['Total_Incidents'].median()
model_data['High_Incidents'] = (model_data['Total_Incidents'] > median_inc).astype(int)
y_clf = model_data['High_Incidents']
```

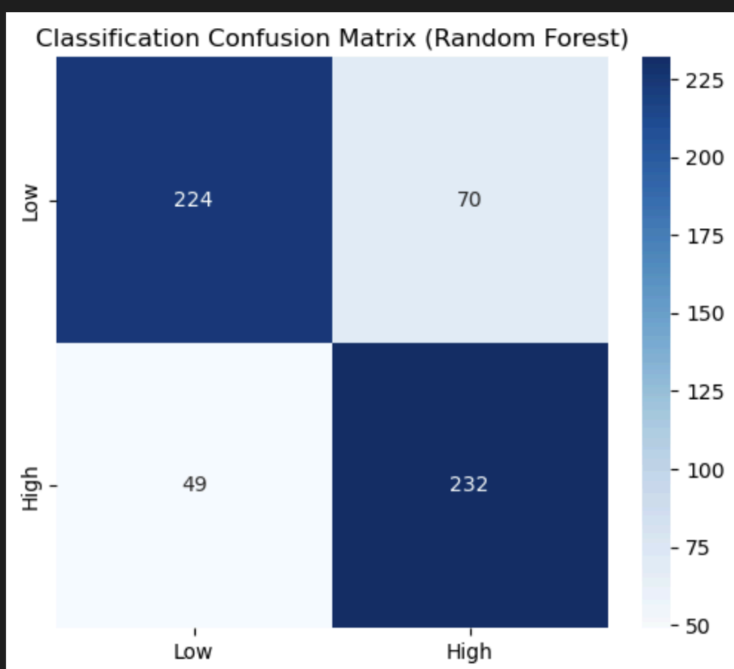
```python
X_train_c, X_test_c, y_train_clf, y_test_clf = train_test_split(X, y_clf, test_size=0.2, random_state=42)

# Method 1: Logistic Regression
log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_c, y_train_clf)
y_pred_log = log_reg.predict(X_test_c)

# Method 2: Random Forest Classifier
rf_clf = RandomForestClassifier(n_estimators=100, random_state=42)
rf_clf.fit(X_train_c, y_train_clf)
y_pred_rf_clf = rf_clf.predict(X_test_c)

# Method 3: k-Nearest Neighbors Classifier
knn_clf = KNeighborsClassifier(n_neighbors=5)
knn_clf.fit(X_train_c, y_train_clf)
y_pred_knn_clf = knn_clf.predict(X_test_c)

# Evaluate Classification
print("\nClassification Results (Accuracy):")
print(f"Logistic Regression: {accuracy_score(y_test_clf, y_pred_log):.4f}")
print(f"Random Forest:       {accuracy_score(y_test_clf, y_pred_rf_clf):.4f}")
print(f"kNN Classifier:      {accuracy_score(y_test_clf, y_pred_knn_clf):.4f}")

# Actual vs Predicted Plot
plt.figure(figsize=(8, 5))
plt.scatter(y_test_reg, y_pred_rf_reg, alpha=0.4)
plt.plot([y_test_reg.min(), y_test_reg.max()], [y_test_reg.min(), y_test_reg.max()], 'r--')
plt.title('Actual vs Predicted Total Incidents (Random Forest)')
plt.xlabel('Actual')
plt.ylabel('Predicted')
plt.savefig('regression_performance.png')

# Feature Importance
importances = pd.Series(rf_reg.feature_importances_, index=features).sort_values()
plt.figure(figsize=(8, 5))
importances.plot(kind='barh', color='teal')
plt.title('Feature Importance for Total Incidents')
plt.savefig('feature_importance.png')

# Confusion Matrix
cm = confusion_matrix(y_test_clf, y_pred_rf_clf)
plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Low', 'High'], yticklabels=['Low', 'High'])
plt.title('Classification Confusion Matrix (Random Forest)')
plt.savefig('classification_cm.png')
```

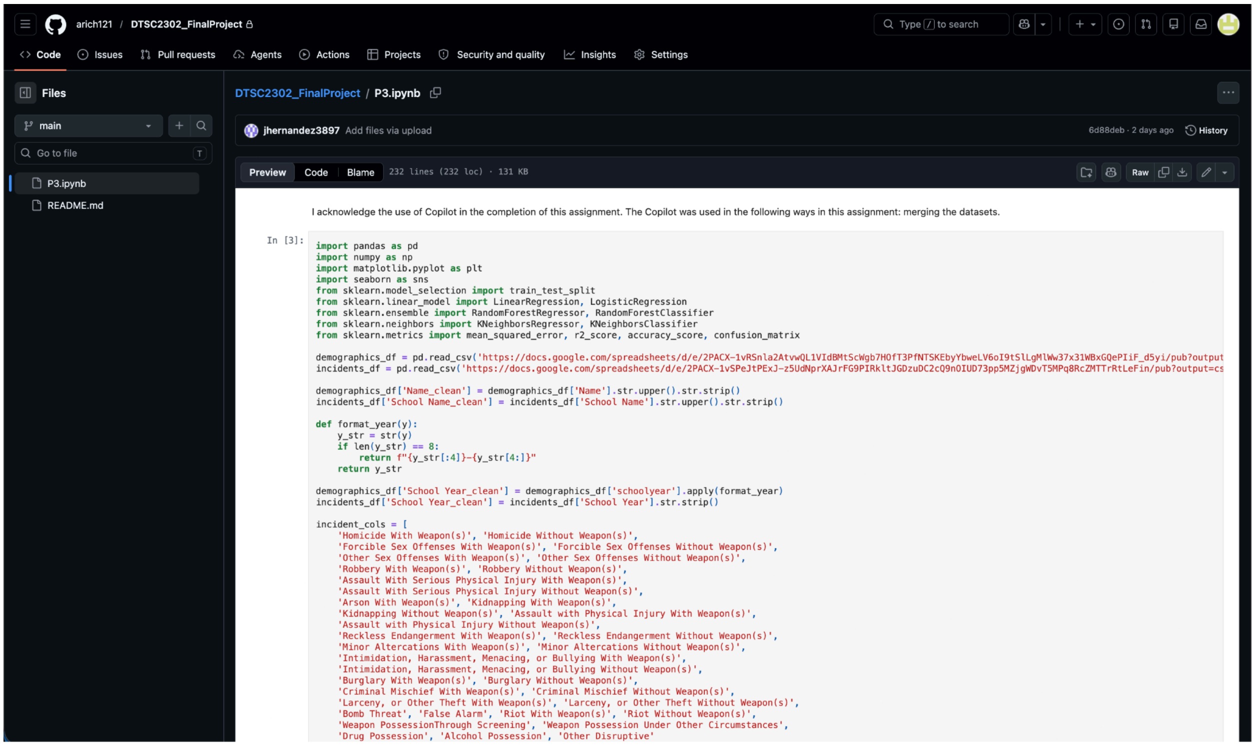
✓ 7.4s

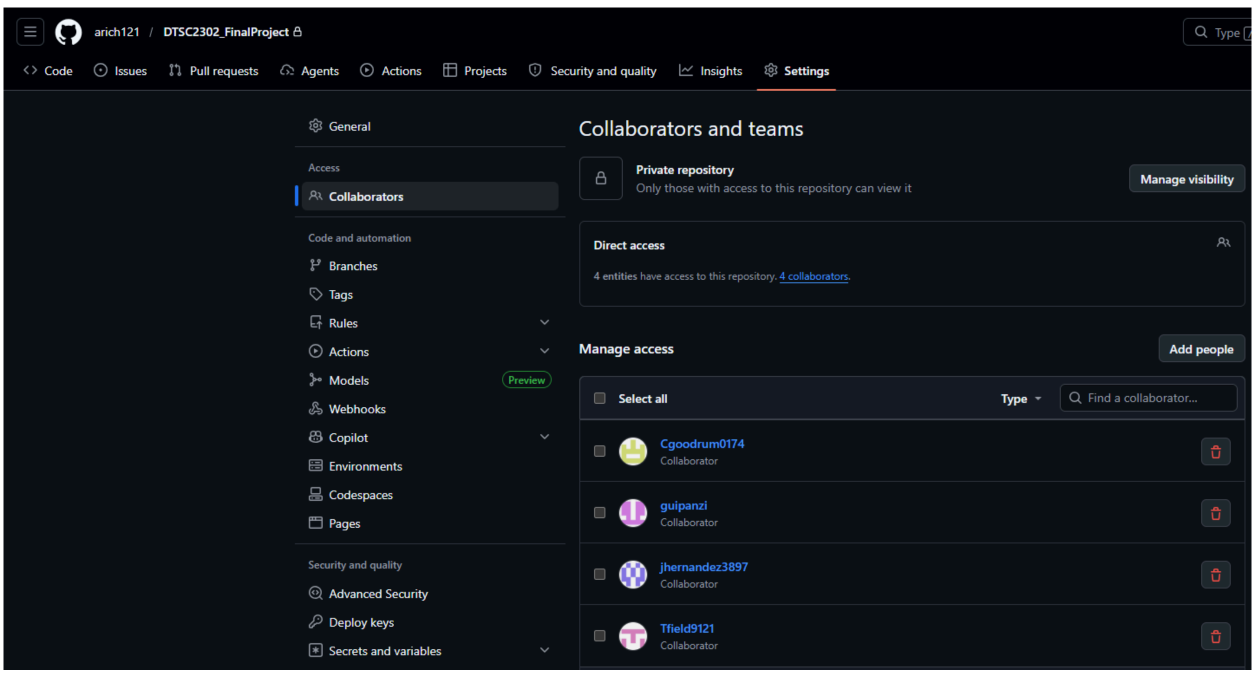
```
Regression Results (R2 Score):
Linear Regression: 0.3014
Random Forest:     0.5214
kNN Regressor:     0.1520

Classification Results (Accuracy):
Logistic Regression: 0.7200
Random Forest:       0.7930
kNN Classifier:      0.6817
```




Feature Importance for Total Incidents
total_enrollment
sped_percent
black_per
white_per
asian_per
male_per
ell_percent
hispanic_per
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Classification Confusion Matrix (Random Forest)
Low
High
224
70
49
232
Low
High
225
200
175
150
125
100
75
50


arich121 / DTSC2302_FinalProject
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Security and quality
Insights
Settings
Files
main
Go to file
P3.ipynb
README.md
DTSC2302_FinalProject / P3.ipynb
jhernandez3897 Add files via upload
6d88deb · 2 days ago
History
Preview
Code
Blame
232 lines (232 loc) · 131 KB
Raw
I acknowledge the use of Copilot in the completion of this assignment. The Copilot was used in the following ways in this assignment: merging the datasets.
In [3]:
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.neighbors import KNeighborsRegressor, KNeighborsClassifier
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, confusion_matrix
demographics_df = pd.read_csv('https://docs.google.com/spreadsheets/d/e/2PACX-1vRSnla2AtvwQL1VIdBMtScWgb7HOfT3PfNTSKEbyYbweLV6oI9tSlLgMlWw37x31WBxGQePIiF_d5yi/pub?output
incidents_df = pd.read_csv('https://docs.google.com/spreadsheets/d/e/2PACX-1vSPeJtPExJ-z5UdNprXAJrFG9PIRkltJGDzuDC2cQ9nOIUD73pp5MZjgWDvT5MPq8RcZMTTrRtLeFin/pub?output=cs
demographics_df['Name_clean'] = demographics_df['Name'].str.upper().str.strip()
incidents_df['School Name_clean'] = incidents_df['School Name'].str.upper().str.strip()
def format_year(y):
y_str = str(y)
if len(y_str) == 8:
return f"{y_str[:4]}-{y_str[4:]}"
return y_str
demographics_df['School Year_clean'] = demographics_df['schoolyear'].apply(format_year)
incidents_df['School Year_clean'] = incidents_df['School Year'].str.strip()
incident_cols = [
'Homicide With Weapon(s)', 'Homicide Without Weapon(s)',
'Forcible Sex Offenses With Weapon(s)', 'Forcible Sex Offenses Without Weapon(s)',
'Other Sex Offenses With Weapon(s)', 'Other Sex Offenses Without Weapon(s)',
'Robbery With Weapon(s)', 'Robbery Without Weapon(s)',
'Assault With Serious Physical Injury With Weapon(s)',
'Assault With Serious Physical Injury Without Weapon(s)',
'Arson With Weapon(s)', 'Kidnapping With Weapon(s)',
'Kidnapping Without Weapon(s)', 'Assault with Physical Injury With Weapon(s)',
'Assault with Physical Injury Without Weapon(s)',
'Reckless Endangerment With Weapon(s)', 'Reckless Endangerment Without Weapon(s)',
'Minor Altercations With Weapon(s)', 'Minor Altercations Without Weapon(s)',
'Intimidation, Harassment, Menacing, or Bullying With Weapon(s)',
'Intimidation, Harassment, Menacing, or Bullying Without Weapon(s)',
'Burglary With Weapon(s)', 'Burglary Without Weapon(s)',
'Criminal Mischief With Weapon(s)', 'Criminal Mischief Without Weapon(s)',
'Larceny, or Other Theft With Weapon(s)', 'Larceny, or Other Theft Without Weapon(s)',
'Bomb Threat', 'False Alarm', 'Riot With Weapon(s)', 'Riot Without Weapon(s)',
'Weapon PossessionThrough Screening', 'Weapon Possession Under Other Circumstances',
'Drug Possession', 'Alcohol Possession', 'Other Disruptive'


arich121 / DTSC2302_FinalProject
Type
Code
Issues
Pull requests
Agents
Actions
Projects
Security and quality
Insights
Settings
General
Access
Collaborators
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
Pages
Security and quality
Advanced Security
Deploy keys
Secrets and variables
Collaborators and teams
Private repository
Only those with access to this repository can view it
Manage visibility
Direct access
4 entities have access to this repository. 4 collaborators.
Manage access
Add people
Select all
Type
Find a collaborator...
Cgoodrum0174
Collaborator
guipanzi
Collaborator
jhernandez3897
Collaborator
Tfield9121
Collaborator